

Introduction to Earth System

- Thermal radiation laws
- Radiative forcing
- EBM-OD

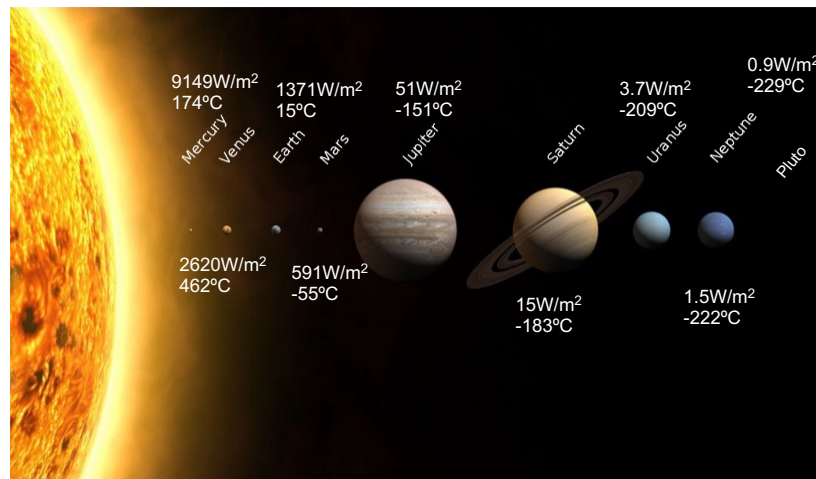
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Solar system

Q: Does the radiation amount a planet receives determine its surface temperature?



Source: wiki, distances not to scale

Answer: It's not only the sunlight

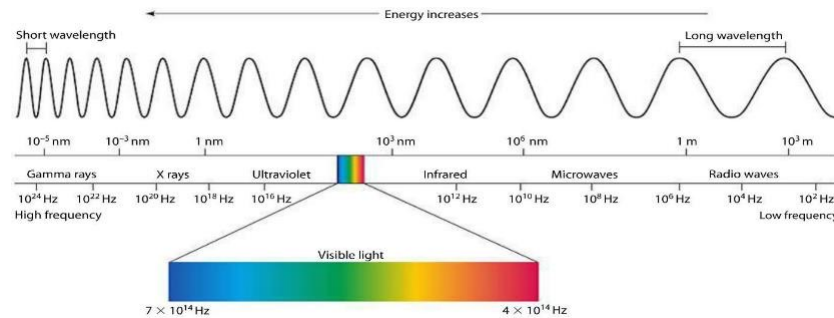
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Some basic definitions

- **Electromagnetic Spectrum:** full range of forms of electromagnetic radiation, which differ by their wavelengths



- Q: In which wave bands do the most common remote sensing systems operate?

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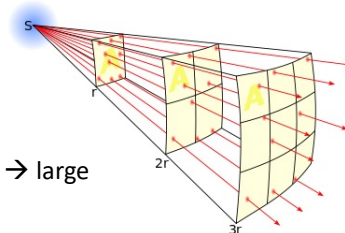
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Some basic definitions

- In electromagnetic radiation, **flux (F)**: amount of energy in an electromagnetic wave that passes **perpendicularly** through a unit surface area per unit time
 - Polar regions are cooler than the tropics
 - Summer temperature are warmer than winter temperature

- Inverse-square law:
(S: solar flux)

$$S = S_0 \left(\frac{r_0}{r} \right)^2$$



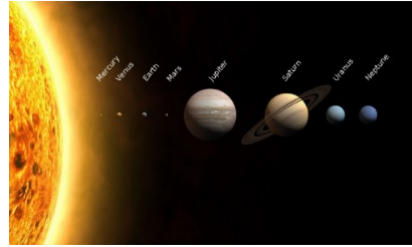
→ Small variations in the shape of Earth's orbit → large changes in the climate of the polar regions

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Q. Inverse-square law

- Venus: an average distance of 0.72 AU from the Sun
- Asteroid belt: orbiting at an average distance of 3.0 AU from the Sun
- Given the average solar flux reaching Earth is: $S_E = 1362 \text{ W/m}^2$



- 1) What is the average solar flux at Venus?
- 2) What is the average solar flux in the Asteroid belt?
- 3) How many times more intense is the sunlight hitting Venus compared to the Asteroid belt?

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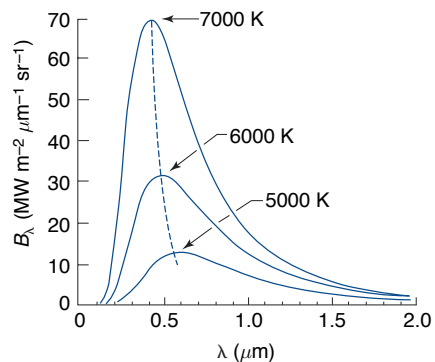
Blackbody radiation

- A **blackbody** is something that completely absorbs all incident radiation
- The radiation emitted by a blackbody is called **blackbody radiation**
- **Planck function:**

$$B_{\lambda}(T) = \frac{c_1 \lambda^{-5}}{\pi (e^{c_2/\lambda T} - 1)}$$

where $B_{\lambda}(T)$ is the intensity of radiation emitted by a blackbody;
 $c_1 = 3.74 \times 10^{-16} \text{ W m}^2$; $c_2 = 1.45 \times 10^{-2} \text{ m K}$

- The Sun emits ~50% of its energy in VIS; ~40% in IR; ~10% in UV



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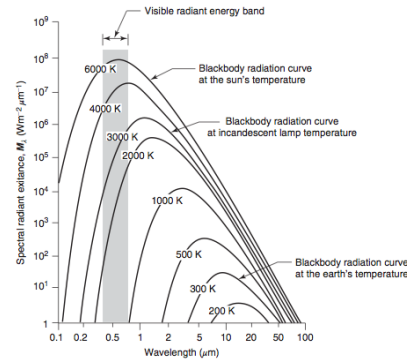
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Wien's law

Wien's displacement law: the wavelength of peak emission for a blackbody at T

$$\lambda_m = \frac{2897}{T}$$

- Prove the Wien's law?
- Q. Peak emission of the Sun (5780K)?
- Q. Peak emission of the Earth (288K)?
- Q. Peak emission of the Cosmic Microwave Background (3K)



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The Stefan-Boltzmann Law

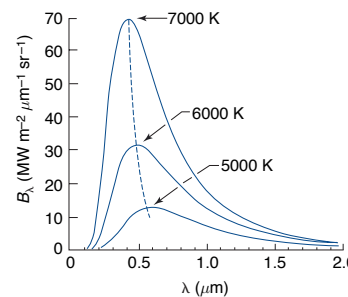
The Stefan-Boltzmann Law: the black body irradiance

$$F = \sigma T^4$$

where σ is the Stefan-Boltzmann constant, equal to $5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$

- For a nonblack body (grey body) $\rightarrow$ estimate T_E
- T_E is called **equivalent blackbody temperature**; or **effective radiating temperature**

Energy emitted from an object is primarily a function of its temperature



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Q. CMB & the Sun

- A satellite equipped with a thermal imaging spectrometer. The sensors are pointed at two different targets in the universe:
 - Target A (the Cosmic Microwave Background): behaves as a near-perfect blackbody at a temperature of **3K**
 - Target B (the Sun): also behaves as a near-perfect blackbody at a temperature of **6000K**
- Q. How many times more total energy per square meter does the Sun emit compared to the CMB?

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Kirchhoff's law

- For an object emitting and absorbing thermal radiation in **thermodynamic equilibrium**, its **emissivity** (ϵ) is **equal** to its **absorptivity** (α) at a given wavelength (λ).

$$\alpha_{\lambda} = \epsilon_{\lambda}$$

→ Corollary?

- ☐ A good/poor absorber is a good/poor emitter
- ☐ An object can emit only at the wavelengths it absorbs
- ☐ An object emits less than a black body of the same size/shape at the same fixed temperature

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Q. Kirchhoff's law

In the infrared spectrum, the atmosphere has an absorptivity of 0.8

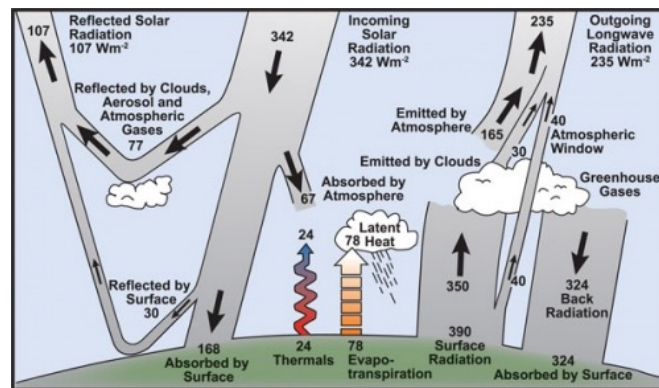
Q. What is the emissivity of the atmosphere in the infrared spectrum?

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Energy balance



- When the climate system reaches a steady state, i.e. temperature stops changing → amount of energy going out = amount of energy coming in
- **Radiative forcing**

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Radiative Forcing

Radiative forcings (IPCC):

- is the change in the net irradiance (Wm^{-2}) at the tropopause
- usually defined as the change relative to the year 1750

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Radiative Forcing

Surface temperature can be linked to radiative forcing:

$$\Delta T_s = \lambda \Delta F$$

where λ is the climate sensitivity ($\text{K}/(\text{W}/\text{m}^2)$)
a typical value of λ is $0.8\text{K}/(\text{W}/\text{m}^2)$

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Radiative Forcing

- Empirical formula

$$\Delta F = \alpha \ln\left(\frac{C}{C_0}\right)$$

- α : a constant, $\sim 5.35 \text{ W/m}^2$
- C : the new concentration of CO_2
- C_0 : the baseline concentration (pre-industrial level is $\sim 280 \text{ ppm}$).

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Q. Estimate the change in today's surface temperature compared to the pre-industrial's temperature, given:

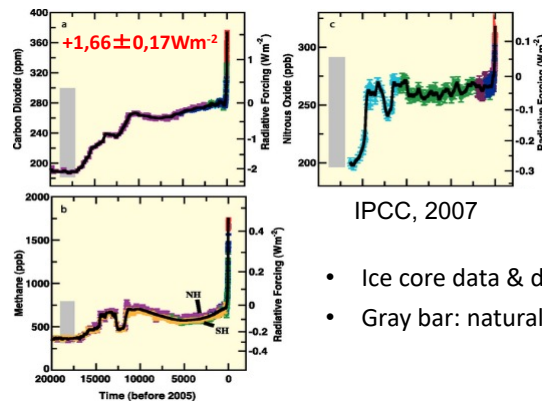
- today's CO_2 concentration: 430 ppm
- pre-industrial level: 280 ppm

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Changes in CO₂, CH₄ & N₂O concentrations



IPCC, 2007

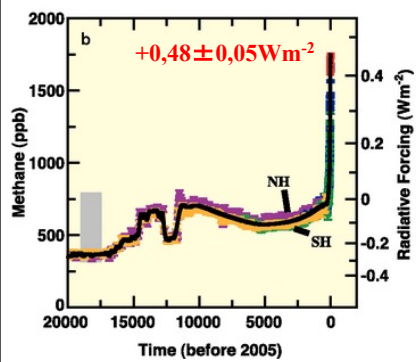
- Ice core data & direct observation (red line)
- Gray bar: natural variability within 650000 yrs

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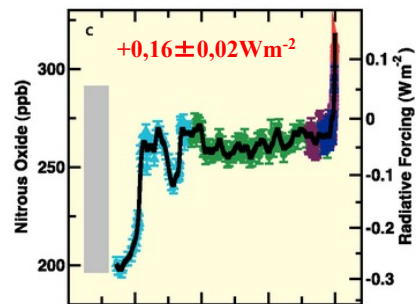
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CH₄



N₂O



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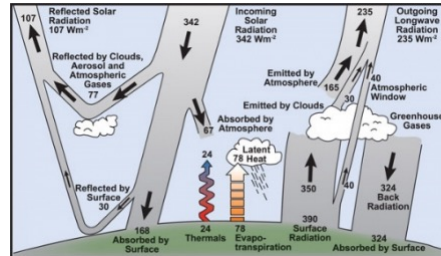
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Energy balance

- Q. Earth's surface temperature generally depends on which factors?

1. Solar flux
2. Earth's reflectivity, called albedo
3. Heat exchanges with the atmosphere
4. Energy radiated by the surface
5. The amount of warming provided by the atmosphere



EBM: Energy Balance Model

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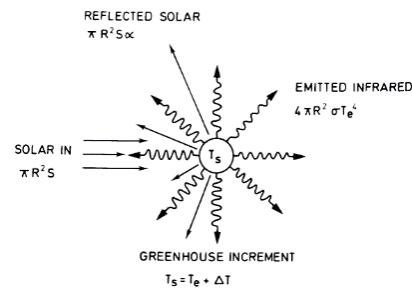
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O-D EBM

Developing equations:

- Incoming Energy
- Outgoing Energy
- Balance



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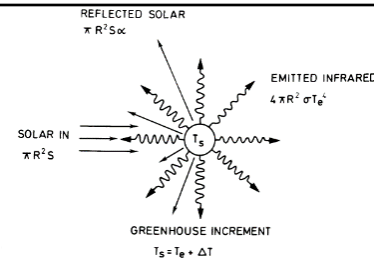
0-D EBM

$$(1 - \alpha)S/4 = \sigma T^4$$

- S: Solar constant, for the Earth, $S=1370\text{W/m}^2$
- α : planetary albedo (~ 0.3)
- σ : Stefan-Boltzman constant ($5.6696\text{E-}8\text{ W/m}^2\text{K}^4$)

→ Black body equivalent temperature:

$$T_e = 255\text{ K (i.e., } -18^\circ\text{C)}$$



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0-D EBM

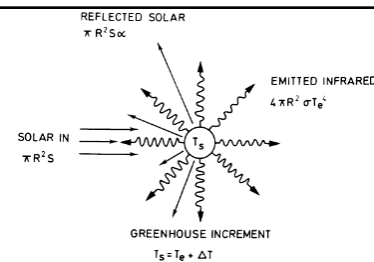
- there are GHGs in the atmosphere
- surface temperature $T_s > T_e$.

$$T_s = T_e + \Delta T$$

where ΔT : greenhouse increment

or $\epsilon \sigma T_s^4 = \sigma T_e^4$

where ϵ : planetary emissivity



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O-D EBM

- For the Earth, $\Delta T \sim 33\text{K}$. Let $\alpha=0.3 \rightarrow ? T_s$
- Let $\epsilon=0.6 \rightarrow ? T_s$

Q. Venus: measured $T_s \sim 730\text{K}$. $S=2619\text{Wm}^{-2}$, $\alpha=0.7 \rightarrow T_e=?$

ϵ replaces a very complex process $\rightarrow$ parameterization

The procedure to find ϵ so that T_s is the closest to the real value $\rightarrow$ **tuning**

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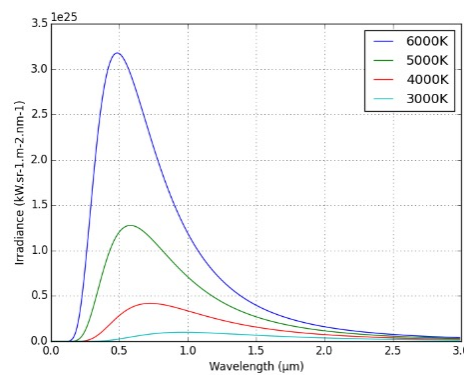
Practice #3.1

1. Plot the Planck curves for different black body temperatures $T=6000\text{K}$, 5000K , 4000K , 3000K
2. Print out the wavelength of peak emission for each body
3. Print out the total energy emitted by each body by using 1) the Stefan-Boltzmann Law, and 2) the Plank function

$$B_\lambda(T) = \frac{c_1 \lambda^{-5}}{\pi (e^{c_2/\lambda T} - 1)}$$

$$c_1 = 3.74 \times 10^{-16} \text{ W m}^2$$

$$c_2 = 1.45 \times 10^{-2} \text{ m K}$$



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Practice #3.2 O-D EBM (cont.)

Given the solar constant of 1368 W/m^2 , $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$ is the Stefan-Boltzmann constant. α is the planetary albedo

- Given the outgoing longwave radiation of the planet by $\epsilon\sigma T_s^4$, where ϵ is the emissivity of the atmosphere, T_s is the surface temperature.
- 1. $\alpha = 0.32$, write a python program to plot the dependence of T_s on ϵ
- 2. $\epsilon = 0.66$, write a python program to plot the dependence of T_s on α
- 3. * Write a python program to plot the dependence of T_s on both α and ϵ